
FROM EVIDENCE TO ACTION: DIAGNOSING SCIENTIFIC AGENTS ACROSS EXPERIMENTATION, LAW FORMATION, AND DECISION TRANSFER

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ABSTRACT

Scientific agents are often evaluated by the best outcome reached during an experimental campaign, conflating evidence acquisition, predictive revision, law formation, and action selection. We study these transformations separately in ChemWorld, a suite of resource-bounded chemical environments with persistent state, complete executable action plans, held-out prediction queries, typed laws, and exact replay. Across 135 public agent sessions spanning entity, parametric, and structural initial-model interventions, agents frequently improved outcomes and predictions within a session, yet the value of a correct prior was task dependent and executable-law fidelity remained limited. Controlled matched-evidence studies showed that direct counterevidence could correct numerical predictions without reliably recovering the registered structural law. In fresh multi-task action selection, only 11 of 42 eligible sessions chose the true best unseen plan. Finally, the oracle control itself failed to align with the decision estimand: a 96-query oracle passed a complete-rank correlation gate in 7 of 8 fresh units but selected the true top action in only 1, whereas a 320-query oracle failed the same gate on its first new world despite selecting the true best action with zero regret. Evidence acquisition, numerical revision, structural identification, action transfer, and evaluator validity are therefore distinct requirements for trustworthy scientific agents.

1 INTRODUCTION

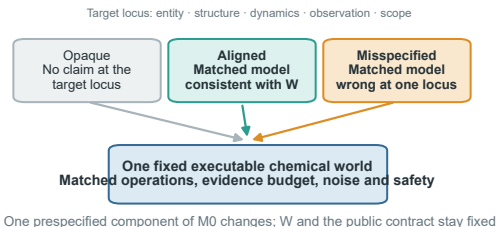
An experimental agent can succeed without learning the right science. It may inherit a correct model, stumble into a productive region, or patch its actions while retaining a false explanation of the world. An endpoint benchmark assigns the same success to all three cases. For a scientific agent, however, they represent different capabilities: prior knowledge, local optimization, and evidence-driven correction. The relevant question is not only whether an agent finds a good experiment, but what that experiment changes in its model of the world.

This distinction matters as language-model agents plan syntheses, call chemistry tools, operate instruments, and enter self-driving laboratory workflows (Boiko et al., 2023; Bran et al., 2024; Szymanski et al., 2023; Darvish et al., 2025; Song et al., 2025; Vriza et al., 2026). Interactive environments increasingly test repeated cycles of hypothesis, intervention, and inference (Jansen et al., 2024; Gandhi et al., 2025; Duan et al., 2025; Zheng et al., 2026; Yang et al., 2026; Batzoglu, 2026). Yet pretrained knowledge, prompt-provided information, experiment selection, endpoint optimization, and verbal explanation are usually observed together. Consequently, a high score need not identify which transformation succeeded, and a low score need not identify which transformation failed.

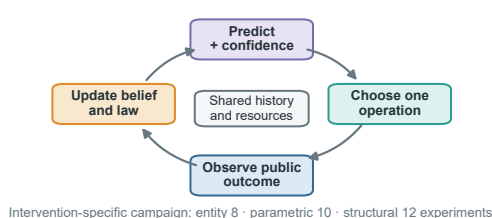
We make those transformations observable by intervening on the agent’s initial world model while holding the executable world fixed. The same hidden world is presented with an opaque, aligned, or misspecified model at a declared entity, parametric, or structural locus. Persistent agents then interact through a transactional laboratory interface. Evaluator-owned counterfactual queries, matched contradictory evidence, executable-law assays, and unseen complete ActionPlans separately measure search, prediction, structural identification, compression, and decision transfer.

Endpoint success does not reveal what the agent learned

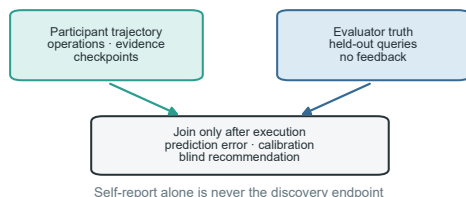
a Intervene on the initial model, not the world



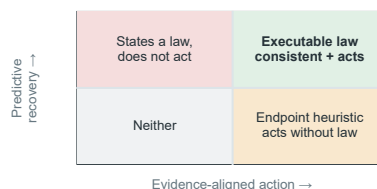
b Trace one evidence-to-action trajectory



c Score understanding after the campaign



d Locate the conversion that failed



The agent-facing initial model is manipulated under one fixed world; search, prediction, executable law and unseen-plan selection remain separate evidence channels.

Figure 1: Endpoint success underdetermines scientific competence. The same outcome can arise from a correct prior, local search, or evidence-driven model correction. ChemWorld intervenes on the initial model while holding the executable world fixed, then evaluates the transitions from evidence to predictions, executable laws, and unseen actions.

Our contribution is an evidence-to-action diagnostic rather than another scalar leaderboard. First, a 135-session prospective programme shows that priors redirect trajectories and that all arms learn predictively, while the registered selective-correction criterion fails at every locus. Second, matched evidence localizes the structural bottleneck: numerical predictions converge near truth, yet the registered power law is recovered in 0/5 misspecified worlds. Third, a 45-cell multi-task assay shows only 11/42 eligible terminal selections at true Top-1. Finally, qualification of a planned causal action-transfer control reveals evaluator misalignment: complete-ranking correlation and decision quality disagree in both directions. The result is a capability chain with empirically distinct failure locations, including failure in the evaluator itself.

2 ENVIRONMENT AND EVALUATION

2.1 FIXED WORLD, PROGRAMMABLE INITIAL MODEL

ChemWorld separates three objects. The executable world contains hidden state, transition and measurement laws, resources, and terminal assays. The public task exposes typed operations, instruments, observations, and an initial model. The evaluator owns held-out truth, replay, and scoring. For world w and initial-model arm a , the hidden transition process remains fixed while the supplied model $M_{0,a}$ changes. The aligned model is correct at one declared locus, the misspecified model is explicit but wrong at that locus, and the opaque arm withholds the corresponding structure. Because operations and measurements remain identical, differences can be attributed to information supplied to the complete agent–tool system rather than to different physics.

Actions are transactions, not free-form prose. The host validates a typed request, checks resources and preconditions, commits the state transition, and returns a public observation or a structured rejection. Failed actions and discards remain in the trajectory. Every submitted action, commit, rollback, measurement, assay, and resource delta is recorded for exact replay. These semantics matter: an action recommendation is scientific evidence only when the evaluator can execute the same complete plan without filling in hidden workflow choices.

Table 1: **Executed evidence layers and claim boundaries.** A completed work package may contain a retained scientific rejection; unstarted units are not silently removed.

Layer	Units	Execution	Supported role
Prospective priors	135 sessions	1,243/1,260 experiments	Search, prediction, law, incumbent replay
Matched evidence	30 sessions	fixed evidence packets	Numerical update versus structural recovery
Open action	45 cells	42 eligible; 240/240 truth/replay	Descriptive unseen-plan selection
96-query control	15 planned	8 attempted; 896/896 truth/replay	Retained pre-participant rejection
320-query control	7 exposed + 15 fresh	7 exposed pass; 1 fresh attempted	Construction repair versus prospective rejection
Gate alignment	16 unit versions	0 new execution	Rank validity versus action validity

2.2 FIVE-STAGE CAPABILITY CHAIN

We distinguish five downstream outcomes. **Evidence acquisition** asks which informative experiments are selected and observed. **Numerical revision** asks whether held-out counterfactual prediction error falls. **Structural identification** asks whether the agent rejects a false form and recovers the registered relation. **Executable compression** asks whether a typed law preserves the information in conditional predictions. **Decision transfer** asks whether the final state selects a good previously unseen plan. A sixth requirement, **evaluator validity**, asks whether the control and metric actually measure the decision estimand. Success at one stage is neither defined as nor assumed to imply success at the next.

At five checkpoints per session, the participant reports initial-model reliability, predictions, uncertainty, evidence references, an executable law, and the next experimental intent. The evaluator executes prespecified counterfactual query sets independently of the participant. It scores 675/675 checkpoints from 420/420 truth executions and later runs each final typed law on the same coordinates. Paired blind replay evaluates the final recommendation against the observed incumbent, while a separate longitudinal assay reveals eight outcome-hidden complete ActionPlans only after autonomous exploration has ended.

3 EXPERIMENTAL PROGRAMME

The prospective programme is layer-stratified. Entity interventions cover five task families and five independently selected public worlds per task. Parametric and structural blocks each cover two validated task families and five worlds. Every task–world cluster contains opaque, aligned, and misspecified arms, yielding 45 clusters and 135 independent persistent sessions. Campaign length is locus-specific: eight, ten, or twelve complete experiments, with five checkpoints in every session. Exploratory, validation, prospective, matched-evidence, and open-action worlds remain separated.

The primary contrast tests selective evidence-driven correction. If $E_{a,k}^{(\ell)}$ is held-out error for arm a , checkpoint k , and locus ℓ , then

$$C_\ell = \left(E_{\text{mis},0}^{(\ell)} - E_{\text{mis},K}^{(\ell)} \right) - \left(E_{\text{aligned},0}^{(\ell)} - E_{\text{aligned},K}^{(\ell)} \right).$$

Success requires greater correction in the misspecified arm, improvement of that arm, and no material deterioration of the aligned arm. Loci are decided separately; unlike intervention semantics are not pooled. Failed scientific cells stay in their scheduled denominator, with the last valid checkpoint carried forward for right-censoring and zero primary improvement assigned when the final prediction is missing. Only a pure infrastructure failure without a persisted trajectory can resume under a fixed attempt cap.

Matched-evidence sessions use cloned worlds and provide the same decisive counterevidence to every arm, separating failure to seek evidence from failure to update after seeing it. The longitudinal action matrix separately contains three tasks, five worlds, and three arms (45 scheduled cells). After 12 autonomous experiments, each agent ranks eight new plans; regret and Top-1 are primary action readouts, while complete-rank correlation and law adequacy are diagnostics. Table~1 keeps these layers and their claim boundaries explicit.

4 RESULTS: FROM PRIORS TO LAWS

4.1 PRIORS REDIRECT SEARCH BUT DO NOT GUARANTEE SELECTIVE CORRECTION

All 135 scheduled sessions produced final records. Participants completed 1,243/1,260 planned experiments; 121 sessions met operational eligibility. The denominator retains 26 discarded lifecycles, 13 resource-ledger rejections, and all right-censored cells. Every session submitted five checkpoints, providing 6,300 counterfactual predictions and 24,300 query–metric values.

The intervention reached behavior. The first complete recipe differed between aligned and misspecified cells in 45/45 matched clusters, between opaque and aligned in 45/45, and between opaque and misspecified in 44/45. This is a manipulation check rather than a causal effect estimate because there are no repeated same-arm sessions. Search continued after the first proposal: 91.2% of completed experiments used a unique recipe, 84.4% of session optima appeared after the midpoint, and 32.6% appeared in the last completed experiment.

Correct-prior utility was task dependent. In entity-level partition, aligned information produced a +0.200 best-endpoint advantage over the misspecified arm in 5/5 worlds. In structural crystallization, a +0.141 first-experiment head start narrowed to +0.055 as the disadvantaged arm explored. In structural partition, aligned and misspecified descriptions both helped relative to opaque identifiers while differing little from one another. A correct model therefore altered the search landscape without imposing one endpoint ordering.

Prediction error nevertheless fell on average in every arm at every locus. Reductions for opaque, aligned, and misspecified cells were 0.111/0.097/0.097 at the entity locus, 0.090/0.033/0.065 at the parametric locus, and 0.219/0.228/0.221 at the structural locus. The stricter selective-correction contrasts were -0.214 for entity ($p = 0.990$), +0.033 for parametric ($p = 0.079$), and -0.224 for structural ($p = 1.000$); none passed. General predictive learning did not become preferential correction of a wrong starting model (Fig.~2).

4.2 MATCHED EVIDENCE SEPARATES NUMERICAL REVISION FROM STRUCTURAL IDENTIFICATION

Free discovery cannot distinguish failure to seek diagnostic evidence from failure to use it. In the parametric matched-evidence block, all five misspecified summaries rejected the supplied high-potential direction and recovered the peak-and-collapse response once the decisive profile was provided. Part of the free-discovery loss therefore lay in evidence acquisition.

The corrected structural assay showed a different bottleneck. After every arm received the same direct phase-process evidence, mean normalized errors fell from 0.2255, 0.2736, and 0.3392 to 0.0074, 0.0060, and 0.0071 for opaque, aligned, and misspecified cells. The misspecified-minus-aligned update contrast was positive in only 3/5 worlds (exact one-sided sign-flip $p = 0.125$), so this is numerical convergence rather than a confirmatory arm effect. Yet 0/5 misspecified public summaries recovered the registered 1.75 power law, 1/5 explicitly rejected the supplied linear form, and 5/5 adopted a saturation or endpoint model. Direct evidence supported accurate local prediction without structural identification (Fig.~3).

All 135 final typed laws executed on their prespecified coordinates, but executability did not preserve the information in conditional predictions. Mean law error was 0.237; relative to effective final predictions, laws were better in 50 cells, equal in one, and worse in 84, with mean law-minus-prediction error +0.069. Typed syntax solved execution coverage, not faithful scientific compression. Paired blind replay likewise tested only incumbent retrieval: 726/726 executions completed for 121 evaluable cells, with recommendations better/equivalent/worse in 1/119/1 cells. Selection beyond the observed campaign required a separate assay.

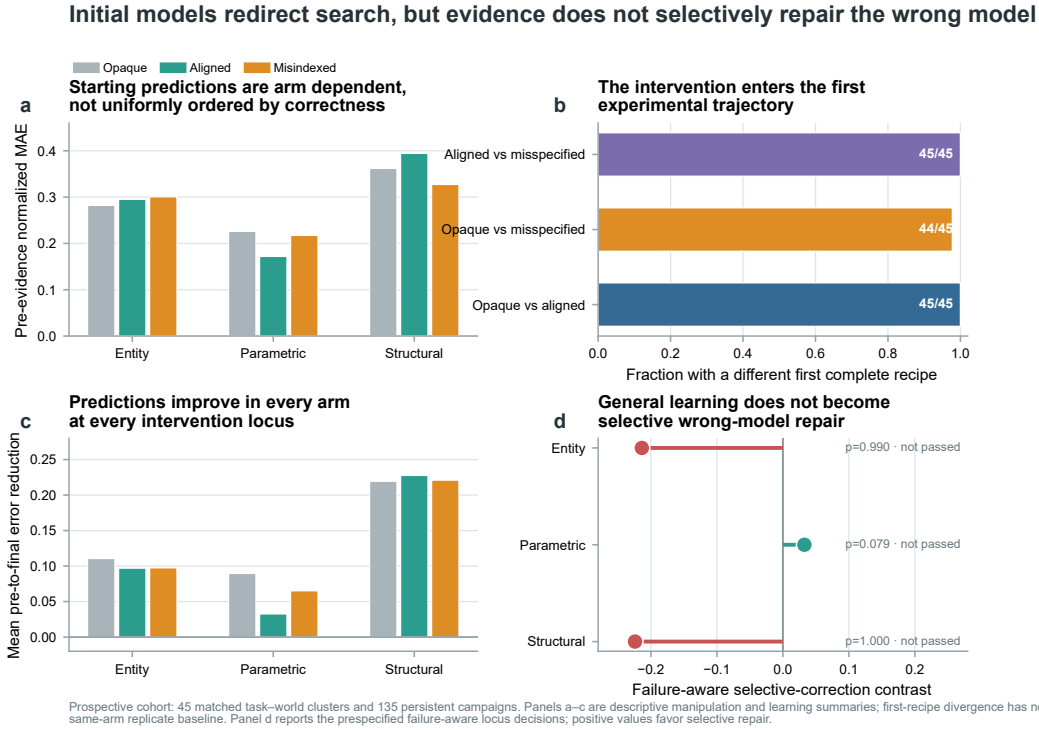


Figure 2: Initial models redirect search, but evidence does not selectively repair the wrong model. Pre-evidence errors, first-recipe divergence, pre-to-final error reductions, and frozen failure-aware selective-correction contrasts are reported by intervention locus. Every arm improves on average, but no locus passes the selective-correction criterion.

5 RESULTS: LAWS TO ACTIONS

5.1 SELECTION AMONG UNSEEN COMPLETE PLANS IS PARTIAL

The longitudinal matrix produced records for 45/45 scheduled cells. Truth evaluation and exact replay both completed 240/240 plan executions. Forty-two cells were uncontaminated and eligible for action metrics; two agent-induced resource/process failures and one session interruption in crystallization remain in the scheduled denominator.

Only 11/42 eligible terminal readouts selected the true Top-1 plan. Mean selected rank was 3.31 of 8 and mean normalized regret was 0.297. The uniform-random rank 4.5 is geometric context, not a no-evidence control, so these outcomes describe post-campaign competence rather than a causal benefit of exploration. Law and action also separated: 30 cells had an inadequate law and wrong action, 11 an inadequate law but correct action, one an adequate law but wrong action, and none an adequate law and correct action. Task means ranged from rank 2.00 for reaction safety to 4.58 for crystallization, and every pairwise arm contrast changed sign under some leave-one-cluster-out omission. We therefore make no pooled arm-level claim (Fig.~4).

5.2 THE ORACLE GATE DOES NOT EQUAL THE ACTION ESTIMAND

A planned five-condition causal follow-up required a provider-free oracle law to rank the eight candidates with Spearman $\rho \geq 0.80$ before any participant was called. The first eight of 15 fresh clusters completed 896/896 truth executions and exact replays. Candidate-opportunity gates passed 8/8, but the eighth oracle obtained $\rho = 0.738095$ and disagreed on Top-1. The frozen stop rule left seven clusters, the operational canary, all 225 participant sessions, and 540 planned participant experiments unstarted. This is a rejection of the control construction, not an estimated participant effect.

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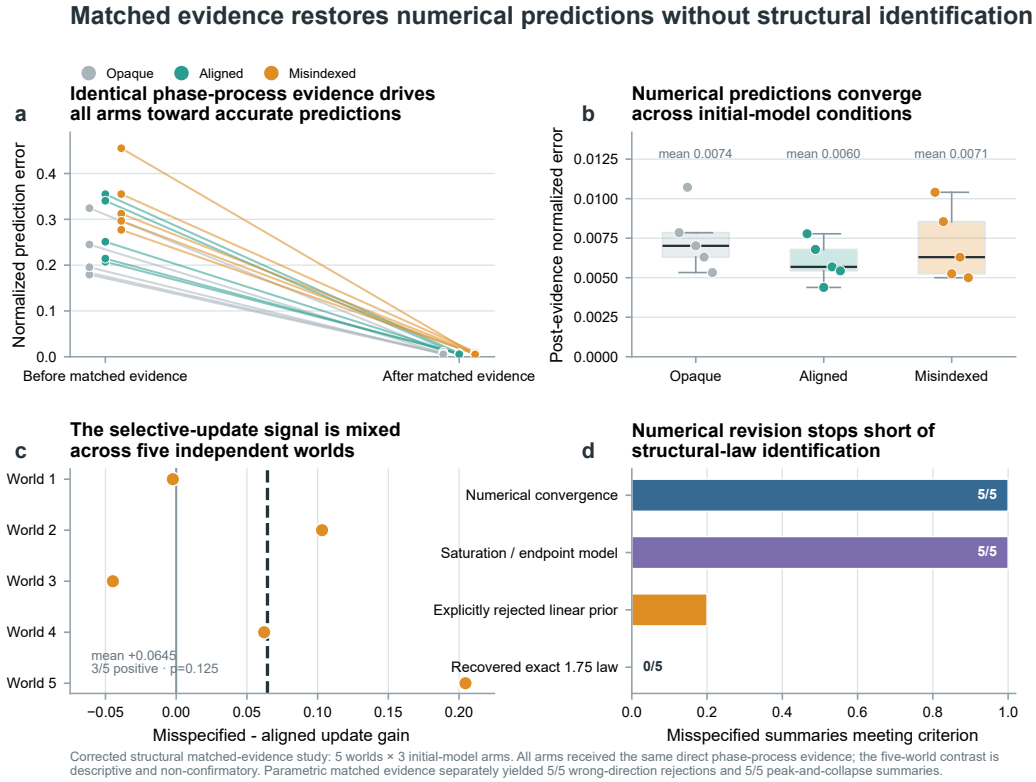


Figure 3: **Matched evidence restores numerical predictions without structural recovery.** All 15 corrected structural cells receive identical evidence and converge near truth. Across the five misspecified worlds, exact recovery of the registered power law is absent despite low post-evidence error.

We then isolated grid coverage while retaining the fitted ExtraTrees family and the correlation threshold. Expanding from 96 queries to 64 global plus 256 candidate-neighborhood queries repaired all four historical failures: 7/7 exposed construction units passed with 2,352/2,352 truth executions and exact replays, and minimum ρ was 0.857143. Prospective validity did not follow. The first new electrochemical world completed 336/336 truth executions and replays but obtained $\rho = 0.714286$, triggering the same stop boundary and leaving 14 fresh clusters unstated. Nevertheless, this oracle selected the true Top-1 action with normalized regret zero.

A frozen zero-execution diagnostic reproduced the original rank and Top-1 outcomes for all 16 completed 96- and 320-query unit versions. Among the eight fresh 96-query units, the rank gate passed 7/8 but Top-1 was correct in only 1/8 and within 0.01 of optimum in 3/8; six were rank-pass/action-wrong. The fresh 320-query unit was rank-fail/action-correct. Thus global-order correlation and decision validity disagree bidirectionally. The historical gates and stops remain valid for their protocols, but future controls must prospectively prioritize regret, near-optimal selection, and near-tie-aware ordering, retaining full-ranking correlation as a secondary diagnostic.

For eight candidates without ties, Spearman correlation is $\rho = 1 - \sum_i d_i^2 / 84$, where d_i is the displacement of candidate i between predicted and true ranks. The retained 96-query failure, $\rho = 0.738095$, therefore represents $\sum_i d_i^2 = 22$; the fresh 320-query failure, $\rho = 0.714286$, represents 24. The scalar says how much the full ordering moved, not where the displacement occurred. The former selected the wrong Top-1 while the latter preserved it exactly, making the estimand mismatch concrete rather than merely threshold dependent.

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Terminal selection of unseen plans is partial and task dependent

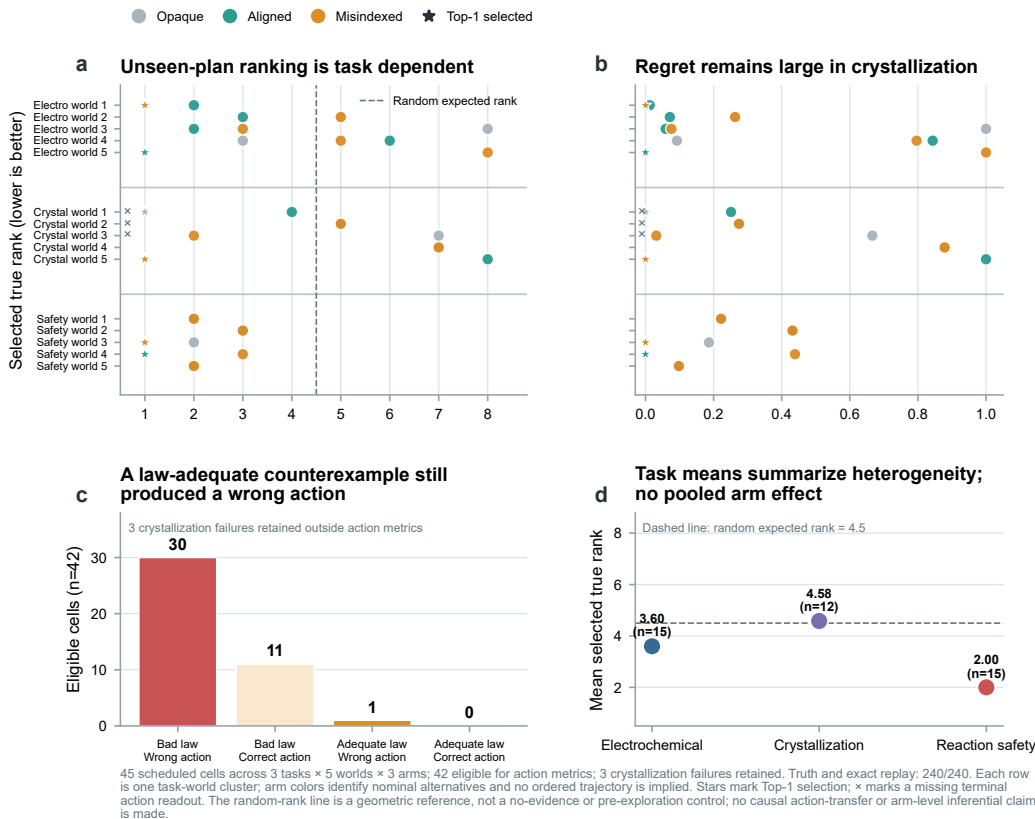


Figure 4: **Terminal selection of unseen plans is partial and task dependent.** Selected true rank, normalized regret, joint law–action categories, and task-level mean ranks are shown for 42 eligible cells. Three retained crystallization failures remain in the 45-cell scheduled denominator. The random-rank line is not an experimental control.

Table 2: **Complete-ranking gates and action endpoints are different estimands.**

Frozen block	<i>n</i>	Rank pass	Top-1	≤ 0.01 regret
Fresh 96-query	8	7	1	3
Exposed 320-query	7	7	4	6
Fresh 320-query	1	0	1	1

6 RELATED WORK

Self-driving laboratories and chemistry agents emphasize closed-loop execution, tool use, or endpoint optimization (Felton et al., 2021; Häse et al., 2021; Boiko et al., 2023; Bran et al., 2024). Virtual laboratories and process-control environments provide scalable interaction and safety (Beeler et al., 2024; Bloor et al., 2024; Malik et al., 2026; Chen et al., 2025). Scientific-discovery benchmarks increasingly test iterative experimentation, causal inference, and transferable knowledge (Jansen et al., 2024; Gandhi et al., 2025; Duan et al., 2025; Yang et al., 2026; Batzoglou, 2026). ChemWorld complements these directions by making the initial model an experimental variable and by evaluating each transformation from evidence to action with evaluator-owned truth and exact replay. The central distinction is diagnostic: endpoint success, prediction learning, law recovery, and action quality are not interchangeable readouts.

7 DISCUSSION AND LIMITATIONS

The evaluated system frequently searches productively and learns numerically, but its transformations lose information at different locations. Correct priors alter trajectories without a general endpoint advantage. Free evidence reduces prediction error without selectively repairing wrong priors. Direct matched evidence can yield near-exact numerical revision while structural recovery fails. Typed laws can execute yet compress predictions poorly, and a thresholded law does not guarantee the correct unseen action. These are not multiple scores for one latent ability; they are distinct capabilities with different interventions and denominators.

The oracle study extends this diagnosis to the harness. A metric can be reproducible and still be misaligned with the intended decision. Spearman correlation rewards ordering across all candidates, whereas the causal follow-up needs a positive control for selected action and decision loss. Stopping before participant execution prevented a weak control from producing an uninterpretable effect. More generally, evaluator validity is part of scientific-agent validity, not a post-hoc implementation detail.

7.1 IMPLICATIONS FOR SCIENTIFIC-AGENT BENCHMARKS

The capability chain suggests three design requirements. First, benchmarks should separate the intervention from the readout. Manipulating prior information, supplying matched evidence, or hiding candidate outcomes answers different questions and therefore requires different independent units. Checkpoint predictions, law execution, and action regret should remain distinct even when a composite score would be convenient. This makes it possible to tell whether a failure arose in acquisition, revision, representation, or transfer.

Second, positive controls should qualify on the endpoint used by the intended contrast. A law can be outcome-disjoint, exactly replayable, and strongly correlated with a full ranking while still choosing the wrong action. Conversely, a decision can be optimal despite moderate errors elsewhere in the ordering. For selection problems, regret and near-optimality should be primary, with full-rank metrics used to diagnose how much additional structure was preserved. Near ties should be declared before execution so that metric choice cannot respond to an unfavorable world.

Third, the evidence record must preserve failed and unstarted units. The 96-query failure is evidence about a control; the 320-query exposed pass is evidence about construction; the first prospective 320-query failure is evidence about generalization. Pooling them would erase the distinction between repairing known cases and succeeding on fresh cases. Likewise, treating stopped participant sessions as zero-valued effects would invent data. Explicit scheduled, eligible, attempted, completed, and unstarted denominators make negative results usable without turning them into claims they cannot support.

7.2 LIMITATIONS AND CONCLUSION

The evidence is deliberately bounded. It concerns one complete agent-tool configuration in simulated chemical worlds, not a cross-model ranking or a claim about laboratory fidelity. Some contrasts have only five independent worlds. The open-action matrix has no no-evidence or pre-exploration action baseline and therefore supports no causal action-transfer effect. The planned five-condition cohort has zero participant data because its control failed before execution. A single-stratum development pilot of reduced conditions ended with the yoked session right-censored after five of six turns and is not used for causal or arm-level inference. Context-reset artifact portability, private-world replication, and cross-provider generalization remain untested.

The practical implication is to evaluate a scientific agent as a sequence of transformations. Each claim should name its intervention, unit, readout, failure rule, and transfer boundary. On this basis, scientific agency is not an endpoint score: it is evidence acquisition, numerical revision, structural identification, executable compression, and decision transfer, qualified by an evaluator that measures the action actually at stake.

AI USE STATEMENT

Generative AI tools assisted software implementation, experiment orchestration, data analysis, visualization code, literature discovery, and drafting and editing of the manuscript. The authors reviewed the study designs and frozen decision rules, inspected retained failures, verified reported denominators and claims against machine-readable records, and tested AI-assisted code before use. The authors take responsibility for the final content, including text, claims, code, and artifacts produced with generative-AI assistance.

REPRODUCIBILITY STATEMENT

The main paper defines the experimental units, interventions, estimands, failure handling, and stop boundaries. The appendix provides the execution and evaluation details needed to reconstruct each evidence layer. Exact action traces, evaluator-owned truth queries, replay checks, complete failure records, and machine-readable summaries will be included in the anonymous supplementary package.

ETHICS STATEMENT

The experiments use simulated chemical environments and no human participants, personal data, or live laboratory actuation. The principal risks are over-interpreting simulated results and treating an evaluator surrogate as the scientific objective; the claim boundaries and retained negative qualification results are reported to mitigate these risks.

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A APPENDIX

A.1 ENVIRONMENT SEMANTICS AND INTERVENTION CONSTRUCTION

An executable world is a stateful transition system with typed operations, measurement functions, finite resources, terminal assays, and an evaluator-only truth surface. A task publishes the operation and instrument contracts while withholding the latent state and registered relations. A scenario binds a world, public task description, resource card, and initial model. Initial-model arms alter one declared information locus without changing the transition process, measurement process, candidate actions, or evaluator truth.

The entity locus changes material identity or ontology. The parametric locus changes a quantitative response model while preserving the declared entities. The structural locus changes the functional form used to describe a mechanism. Aligned and misspecified models are matched in explicitness so that a contrast is not merely information versus no information. The opaque arm exposes the same laboratory interface but withholds the intervened model component. Exploratory, validation, prospective, and private world selections are disjoint by construction.

A.2 PERSISTENT EXECUTION AND CHECKPOINTS

Each participant cell is one persistent agent process. At every turn, the agent may call only the public laboratory tools. The host validates schemas and preconditions, applies resource accounting, commits successful transactions, and returns public observations. It does not choose or repair an experiment on the participant’s behalf. Entity, parametric, and structural campaigns permit eight, ten, and twelve complete experiments, respectively. Checkpoints occur at experiments 0/2/4/6/8, 0/2/4/7/10, and 0/3/6/9/12.

Every checkpoint records model-reliability judgements, query predictions, uncertainty, public evidence references, a typed executable law, and the intended next experiment. Checkpoint repair is part of the evaluated interface burden. All eventually yielded valid records, but rejected attempts remain in the process log and resource accounting.

A.3 EVALUATOR-OWNED PREDICTION AND LAW ASSAYS

Entity checkpoints contain four prespecified counterfactual queries; parametric and structural checkpoints contain sixteen. The evaluator executes each unique task–world query independently, never returns its truth to the participant, and shares truth across prior arms and checkpoints. The primary readout is mean normalized absolute error across registered query–metric pairs. The executed cohort contains 420/420 truth executions, 1,620 query–metric truth values, 675/675 checkpoints, 6,300 predictions, and 24,300 scored values.

Final typed laws are executed on the same registered coordinates. Schema validity, coordinate coverage, truth-normalized error, change from pre-evidence and effective-final checkpoints, and consistency with explicit final predictions are retained separately. This assay establishes executability and in-domain compression only. Reusability would require new coordinates or transfer to a context-reset process.

A.4 FAILURE-AWARE INFERENCE

The unit of prospective inference is the independently selected task–world cluster, not a checkpoint, query, experiment, or replay. The three prior arms within a cluster are matched. Locus-specific selective-correction contrasts are computed first and are not pooled across different intervention semantics. Failed scientific cells remain assigned. A right-censored session carries its last valid checkpoint forward, and a missing final prediction receives zero primary improvement. Infrastructure recovery is allowed only when no scientific trajectory has persisted and only under the fixed attempt limit.

The matched-evidence structural contrast has five independent worlds. Its interval and exact one-sided sign-flip result are descriptive because the block was designed to localize a transition, not to establish a population-level arm effect. Recipe divergence is likewise a manipulation check: without

repeated same-arm sessions, provider stochasticity cannot be separated from intervention sensitivity at exact-recipe resolution.

A.5 COMPLETE-PLAN ACTION ASSAY

In each open-action cell, a persistent agent performs twelve autonomous experiments and commits checkpoints after 0, 3, 6, 9, and 12 experiments. The host then reveals eight unseen ActionPlans. Each plan specifies ordered operations, all submitted parameters, initial-state assumptions, intermediate measurement positions, terminal assay, and declared omitted operations. Candidate outcomes, ranks, and other-arm evidence stay hidden. The evaluator verifies identity of disclosed, truth-evaluated, and replayed plans.

The primary endpoint is normalized regret of the selected plan within its task–world candidate set. Selected rank, Top-1, pairwise ordering, and law adequacy are secondary or diagnostic. Forty-five cells were scheduled across three tasks, five worlds, and three initial-model arms. Forty-two were eligible. The two agent-induced resource/process failures and one provider interruption are retained as failures in the scheduled denominator and are not replaced. An independent repair is reported only as sensitivity evidence because it follows a new trajectory.

A.6 ORACLE-CONTROL QUALIFICATION

The unexecuted causal follow-up crossed three tasks, five worlds, three priors, and five conditions: no evidence, stepwise yoked evidence, autonomous exploration, learned law in a fresh context, and a provider-free oracle law. The planned denominator was 225 participant sessions, 45 autonomous donors, and 540 donor experiments. Participant execution required every fresh task–world oracle to pass candidate opportunity checks and to attain Spearman rank correlation at least 0.80 over the eight candidate plans using fit data disjoint from candidate outcomes.

The 96-query qualification stopped after eight of fifteen clusters. It completed 896/896 truth executions and exact replays, with eight candidate gates and seven oracle rank gates passing. The retained crystallization failure had $\rho = 0.738095$, Top-1 disagreement, and zero fit/candidate overlap. No operational canary or participant session began.

The 320-query construction held the fitted ExtraTrees family fixed while expanding coverage to 64 global and 256 candidate-neighborhood queries. A platform-defective partial caused by an evidence-reference schema limit was retained and not reused. After the contract repair, a new root completed 2,352/2,352 truth executions and replays across seven exposed units, passing all seven and repairing four historical failures. The first prospective world then completed 336/336 executions and replays but failed at $\rho = 0.714286$; the remaining fourteen were not started. Candidate outcomes were never read during fitting.

The gate-alignment analysis added no execution. It re-read sixteen frozen unit versions and exactly reproduced their original Spearman and Top-1 values. Its action endpoints were Top-1, normalized regret, selection within 0.01 of optimum, and near-tie-aware pair ordering. It did not alter any historical result, threshold, or stop decision.

A.7 ADDITIONAL SCOPE BOUNDARIES

The study evaluates simulated chemistry and one complete agent–tool system. It does not establish physical laboratory fidelity, universal chemical coverage, cross-provider ordering, or artifact portability. The public cohort is not a private-world confirmation. The 45-cell action matrix lacks a same-agent no-evidence baseline. The five-condition causal cohort is unexecuted. A later reduced development pilot completed no-evidence and learned-law sessions in one electrochemical stratum, but its yoked session was right-censored after five of six provider turns; it is excluded from all main causal and arm-level conclusions.